

Assignment 1 Q3: Object Detection: Mathematical Formulation

23K-0842 Kinza

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Introduction

Object detection is a core task in computer vision where the goal is to identify and localize objects in an image. Unlike simple classification, object detection predicts both the **class** of the object and its **location** using bounding boxes.

Formally, given an input image I , a detection model outputs:

$$\{(b_i, c_i)\}_{i=1}^N$$

where:

- b_i = bounding box coordinates of object i
- c_i = class label of object i
- N = number of objects in the image

Bounding boxes can be represented as:

- Center-based: $b_i = (x, y, w, h)$
- Corner-based: $b_i = (x_{\min}, y_{\min}, x_{\max}, y_{\max})$

Loss Functions

The total loss combines classification and localization:

$$L = L_{\text{cls}} + \lambda L_{\text{loc}}$$

- L_{cls} = classification loss - L_{loc} = localization loss - λ = weight balancing the two losses

Classification Loss

Cross-entropy loss is commonly used:

$$L_{\text{cls}} = - \sum_{k=1}^C y_k \log(p_k)$$

Localization Loss

Smooth L1 loss (Huber loss):

$$L_{\text{loc}} = \begin{cases} 0.5(\hat{b} - b)^2 & |\hat{b} - b| < 1 \\ |\hat{b} - b| - 0.5 & \text{otherwise} \end{cases}$$

Intersection over Union (IoU)

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$

- IoU > 0.5 usually considered a correct detection. - IoU is also used in **loss functions** to improve localization.

Anchor Boxes and Regression

Anchor boxes are predefined boxes used to handle multiple object sizes. The model predicts offsets:

$$\delta_x = \frac{x - x_a}{w_a}, \quad \delta_y = \frac{y - y_a}{h_a}, \quad \delta_w = \log \frac{w}{w_a}, \quad \delta_h = \log \frac{h}{h_a}$$

Evaluation Metrics

- Precision = $\frac{TP}{TP+FP}$
- Recall = $\frac{TP}{TP+FN}$
- Detection Error Rate (DER) = $\frac{FP+FN}{N_{\text{detections}}}$

Advanced Topics in Object Detection

1. Multi-Scale Detection

Objects in images can have various sizes. Multi-scale detection uses feature maps at different resolutions:

- High-resolution maps detect small objects - Low-resolution maps detect large objects
- Mathematically, let F_l be a feature map at level l . Predicted boxes at each level:

$$b_i^l = f(F_l)$$

Example: Feature pyramid networks (FPN) combine multiple scales to detect objects of varying sizes.

2. Anchor-Free Detection

Some modern models avoid anchors, directly predicting keypoints or center points of objects.

- Let (c_x, c_y) = predicted center of object - Let (w, h) = predicted width and height

Bounding box:

$$b = (c_x - w/2, c_y - h/2, c_x + w/2, c_y + h/2)$$

Anchor-free detection reduces computation and avoids manually designing anchor boxes.

3. IoU-based Loss Improvements

IoU loss is improved to penalize poor overlap better.

3.1 Generalized IoU (GIoU)

$$L_{\text{GIoU}} = 1 - \text{IoU} + \frac{|C - (A \cup B)|}{|C|}$$

- A = predicted box, B = ground-truth box - C = smallest enclosing box of A and B

3.2 Distance IoU (DIoU)

DIoU adds penalty for center distance:

$$L_{\text{DIoU}} = 1 - \text{IoU} + \frac{\|b_{\text{pred}}^{\text{center}} - b_{\text{gt}}^{\text{center}}\|^2}{c^2}$$

3.3 Complete IoU (CIoU)

CIoU also considers aspect ratio:

$$L_{\text{CIoU}} = L_{\text{DIoU}} + \alpha v, \quad v = \frac{4}{\pi^2} \left(\arctan \frac{w_{\text{gt}}}{h_{\text{gt}}} - \arctan \frac{w_{\text{pred}}}{h_{\text{pred}}} \right)^2$$

- α = trade-off parameter - Penalizes predicted box with incorrect aspect ratio

Example Diagram

